Course Name: Engineering Chemistry Lab
Course Code: CY-102

Contact Hours/Week: 2P **Course Credits:** 01

Course Objectives:

- To analyse water samples for different parameters like amount of residual chlorine, alkalinity and hardness
- To measure physical properties of liquids and systems of different components
- To estimate the percentage of a particular metal in its ore or alloy
- To familiarize students about the characterization method like absorption spectroscopy and adsorption processes
- To determine quantitative composition of a solution through conductometric and potentiometric titration

List of Experiments

1. Estimation of residual Chlorine in a given sample of water
2. Estimation of concentration of hydroxyl, carbonate, bicarbonate and total alkalinity in a given sample of water
3. Estimation of Hardness (Temporary and Permanent) in a given sample of water
4. Determination of quantity of Ferrous ions in a sample of water by KMnO4 titration
5. Estimation of Cu in a given sample of brass by (i) titration and/or (ii) Atomic absorption spectroscopy method
6. Determination of Viscosity of unknown liquid by Ostwald's viscometer
7. Determination of Viscosity using Redwood viscometer.
8. Determination of surface tension of unknown liquid by drop number method.
9. Verification of the Beer-Lambert law by using Colorimetric method and its application in determination concentration of unknown solution.
10. Conductometric titration of a weak acid with strong base.
11. Conductometric titration of a weak acid with weak base.
12. Potentiometric titration of weak acid with strong base.
13. Potentiometric titration of mixture of strong and weak acid with strong base.
14. Determination of flash and fire point of a given bituminous material.
15. Determination of phase diagram of two component system
16. Adsorption study on charcoal.

Note: The concerned Course Coordinator will prepare the actual list of experiments/problems at the start of semester based on above generic list.

Course Outcomes

Upon successful completion of the course, the students will be able to

CO1: Quantify different pollutants in water samples
CO2: Identify the unknown liquid from their surface tension and viscosity measurement.
CO3: Analytically measure the composition
CO4: Understand quantitative analysis of acids and bases
CO5: Understand to describe systems of one/two component